

Applied physics:

Before physics, people studied natural philosophy.

Some philosophers are:

Galileo galilei, Aristotle:

Then natural physics divided into:

① Biological science:

② physical science

Study of living things → Biological sciences:

Study of non-living things → physical sciences:

Physical science again divided into:

① Geology (study of earth)

② Astronomy (study of stars)

③ Chemistry (study of chemicals)

④ Physics:

Physics:

Branch of science which deals with the study of matter and energy & their mutual interaction.

Matter:

Anything that occupy space & have mass:

- Solid
- Liquid
- mass

Energy:

Anything that does not occupy space nor have mass:

- Sound
- heat
- Light

So physics is basically the study of universe:

interaction means $E=mc^2$

Atom bomb → used uranium (matter) when blast (energy) produced

Pair-Production: → Basically energy converts into matter: [High energy photon interact with magnetic field create a particle-antiparticle pair.]

Major physics divided into:

1) Classical physics (Before 17th century)

2) Modern physics (19/20th century)

Modern physics: → Black-Body radiation
→ Max. Planck
→ Einstein

Modern physics also contain:

- nuclear physics
- Quantum physics (Quantum mechanics)
- particle physics

Electro-magnetism:

- Electricity (charge)
- Magnetism (magnets)

Electricity:

Electricity is a phenomenon arises due to the motion of electrical charges (current, electrical current)

Magnetism:

Magnetism is a process due to which materials attract or repel metallic substances, having north and south pole:

→ Maxwell combined electricity and magnetism.

→ Maxwell's equations are four:

These equations are relationship b/w electric field and magnetic field:

Electrical charge:

Electrical charge is intrinsic property of matter associated with its elementary particles, due to which it experience and produce electrical-magnetic effects.

→ Mass is also intrinsic property.

($w = mg$ = weight)

→ The difference b/w mass and charge are:

→ Mass can exist without charge

→ Charge can not exist without mass

→ Neutron = $n^0 = 1.67 \times 10^{-27} \text{ kg}$

→ electron = $1.6 \times 10^{-19} \text{ C}$, mass = $9.1 \times 10^{-31} \text{ kg}$
Proton = $1.6 \times 10^{-19} \text{ C}$, mass = $1.67 \times 10^{-27} \text{ kg}$

Types of Charges:

Benjamin Franklin divided charge into two types:

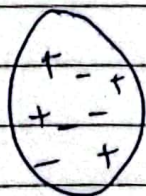
- 1) Positive (Proton)
- 2) Negative (electron)

If we want to charge something negatively, we provide electron (add electrons) accordingly.

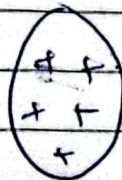
So negative charge is produced by gain of electron
→ To charge positively, remove electrons (b/c adding proton or removing them from nucleus is difficult b/c of strong nuclear force, so positive charge is basically the loss of electrons.

Basic law of electrostatics:

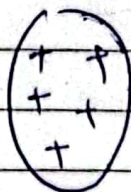
- ① Like charges repel each other.
- ② Opposite charges attract each other.
- ③ A charge body always attracts a neutral body



net charge = 0



positively charged



This is called induction (dipole)
(There will be never repulsion between them b/c neutral body never charged body but they are balanced)

There is other small charges called real charges.

→ They are dimensionless charges.

→ When distance b/w two charges is very large as compared to their sizes, then these charges are called point charges.

→ Charge is scalar quantity.

→ unit is (C) coulombs

$$Q = I \cdot t \text{ (current} \cdot \text{time)}$$

$$C = A \cdot s \text{ (Ampere Second)}$$

~~25000~~ 0.5

$\frac{1}{5000}$

Smallest charges:

Charge on electron & proton is the smallest charges that can exist independently,

$$e \rightarrow 1.6 \times 10^{-19} \rightarrow e^-$$

$$p \rightarrow 1.6 \times 10^{-19} \rightarrow e^+$$

$|e| \rightarrow$ smallest charges:

$e/2 \rightarrow$ does not exist (half electron)

$e/3 \rightarrow$ does not exist

Quarks are 6:

$$\text{up} \rightarrow +2/3 e$$

$$\text{down} \rightarrow -1/3 e$$

$$\text{Strange} \rightarrow -1/3 e$$

$$\text{Charm} \rightarrow +2/3 e$$

$$\text{top} \rightarrow +2/3 e$$

$$\text{Bottom} \rightarrow -1/3 e$$

Because quarks can't be independent, they

will be in pairs:

$1e^-, 2e^-, 3e^-, -1e, -2e \rightarrow$ can exist independently:

Properties of charges:

Charge has three (3) properties:

(1) Additivity of Charge:

Consider a system of charges having three point charges with magnitude q_1, q_2 & q_3 .

Then total charge can be calculated as:

$$Q_T = q_1 + q_2 + q_3$$

If system consist of n , number of charges, then total charge can be written as:

$$Q_T = q_1 + q_2 + \dots + q_n$$

2) Conservation of charges:

Total charge of an isolated system remains constant over time: OR

Algebraic sum of charges in a system remains constant

(ایک ایسا سسٹم جس پر اکسٹرنل (external) فوर्स نہ ہو، isolated) $\Sigma q = \text{constant}$

$$n^0 \rightarrow e^- + e^+$$

$$\Sigma q = \text{constant}$$

OR

$$q_i = q_f \quad (\text{initial is equal to final})$$

In this initial net charge is equal to final net

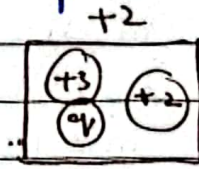
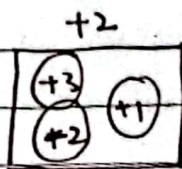
Charge:

$e^- + e^+ \rightarrow \gamma + \gamma$ (Example of law of conservation of energy)

Charge of system remains zero, before and after an

event:

(3) Quantization of charges:



$$+3 + 1 - 2 = +3 + 2 + q$$

$$-1 = +2 + q$$

$$-1 - 2 = q$$

$$-3 = q$$

Quantization means anything that have fin value:

$\left[\begin{array}{l} (1, 2, 3, 4) \text{ discrete} \\ (1, 1.1, 1.2 \dots) \text{ continuous} \end{array} \right]$

Quantization of charge means that an object can

only have charge multiple of "e".

$$+1e, +2e, +3e, \dots$$

یونک e کے loss یا gain کی وجہ سے خارج ہوتا ہے تو ایک الیکٹران یا پروٹون loss ہو سکتا ہے

(gain یا loss)

When we transfer one electron electric charge

from one body to another, then the current flows by electric charges will not be continuous, it has only discrete value:

→ We cannot divide charge into small parts:

$$q = ne$$

$$n = \pm 1, \pm 2, \pm 3, \pm 4, \dots$$

It means that electrical charge can exist only in quantities that are integral multiples of certain elementary quantities of charge 'e'.

$$1C = 1.6 \times 10^{-19}$$

→ How many electrons are in 1C?

$$q = ne$$

$$n = q/e$$

$$n = 1C / 1.6 \times 10^{-19}$$

$$n = 1 / 1.6 \times 10^{19}$$

$$n = 0.625 \times 10^{19}$$

$$\boxed{n = 6.25 \times 10^{18} e} \quad (\text{These electrons are in } 1C)$$

Methods to charge a body:

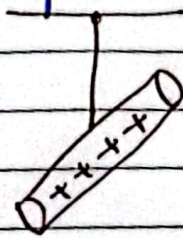
① By friction (Rubbing)

② Conduction

③ induction

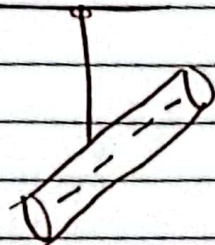
① By friction:

①



Glass rod
Silk

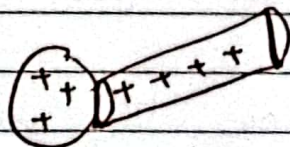
②



Plastic rod
Fur

means when two neutral bodies are rubbed together.

② By conduction:



(in contact)

When a neutral body brought in contact with a charged body:

③ By induction:

When a charged body brought near to neutral body without contact.

Coulombs law:

Quantitative analysis of charges $\rightarrow$ 1st gave Coulombs

$\rightarrow$ It gives quantitative law:

$\rightarrow$ Empirical law (deduced from experiments)

$\rightarrow$ Coulombs Torsion balance (assignment)

$\rightarrow$ also called inverse square law:

$\rightarrow$ 1784

$\rightarrow$ Charles Coulombs

$\rightarrow$ French military engineer:

Statement:

Electrostatic force of attraction or repulsion b/w

two point charges is directly proportional to product

of magnitude of charge and inversely proportional

to the square of distance b/w them:

It is only applicable on point charges;

$\rightarrow$ it is only applicable on static charges

$$F \propto \frac{|q_1| |q_2|}{r^2} \quad \text{--- (A)}$$

$$F = \frac{k |q_1| |q_2|}{r^2} \quad \text{--- (B)}$$

K = Coulombs constant

$$K = \frac{F r^2}{q_1 q_2}$$

$$K = \frac{Nm^2}{C^2} \quad (\text{Newton meter square per coulomb square})$$

Dependency:

① Nature of medium

② System of units:

$$K = \frac{1}{4\pi \epsilon_0}$$

$\epsilon_0 \rightarrow$ Epsilon not
OR
Permittivity of
medium

* In vacuum

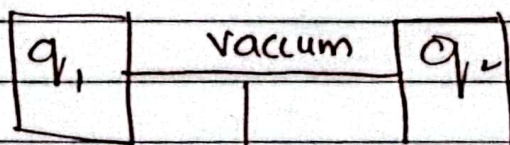
$$F_{vac} = \frac{K_{vac} |q_1| |q_2|}{r^2}$$

$$K = \frac{1}{4\pi \epsilon_0} \quad (\text{Basically } \epsilon_0 \text{ changes})$$

$$\text{For vacuum } \epsilon_0 = 8.85 \times 10^{-12}$$

Unit of ϵ_0 = Newton meter coulombs²

$$K_{vac} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$$

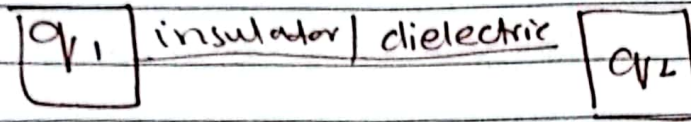


$\rightarrow$ permittivity (permission to attract or repel)

Permittivity $\epsilon' \rightarrow$ permission ϵ'_j

Permittivity $\propto \frac{1}{\text{Permission}}$

→ Coulombs force in medium



$$F_{\text{med}} = K_{\text{med}} \frac{|q_1| |q_2|}{r^2}$$

$$\rightarrow K_{\text{med}} = \frac{1}{4\pi \epsilon_{\text{med}}}$$

$$\rightarrow K_{\text{med}} = \frac{1}{4\pi \epsilon_0 \epsilon_r}$$

$\epsilon_r = \text{relative permittivity}$

$$\rightarrow F_{\text{med}} = \frac{1}{4\pi \epsilon_0 \epsilon_r} \frac{q_1 q_2}{r^2}$$

$\epsilon_r \rightarrow \text{vacuum}$
 $\epsilon_r \rightarrow \text{any medium}$

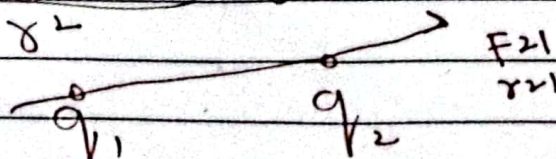
$$\rightarrow F_{\text{med}} = \frac{F_{\text{vac}}}{\epsilon_r}$$

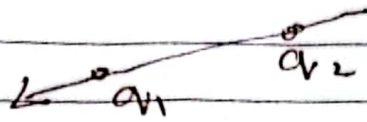
$$\Rightarrow F_{\text{med}} < f_{\text{vac}}$$

→ Presence of di-electric or insulating medium reduces coulombs force by ϵ_r .

Vector form of coulomb's law:

$$\vec{F} = K \frac{|q_1| |q_2|}{r^2} \hat{r}$$





$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r_{12}^2} \vec{r}_{12}$$

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r_{12}^2} \vec{r}_{12}$$

According to Newton's 2nd law,

$$F_{12} = F_{21}$$

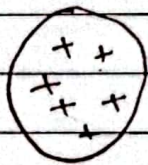
$$\frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r_{12}^2} \vec{r}_{12} = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r_{21}^2} \vec{r}_{21}$$

$$\vec{r}_{12} = -\vec{r}_{21}$$

Charge Distribution

① Uniform Distribution:

→ Continuous distribution



charge distribution is equal in all parts.

It consist of three types:

① Linear charge Distribution:

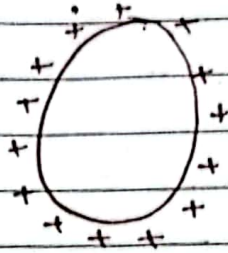
② Surface charge distribution.

③ Volume charge distribution.

Linear charge distribution

↳ Charge distribution in one dimension:

+ + + + + + +

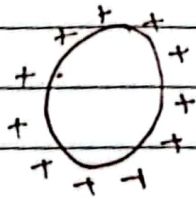


Also called linear charge density (λ)

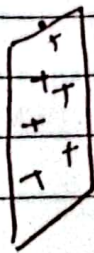
- Charge per unit length is called linear charge distribution:

$$\lambda = \frac{q}{L} \quad \left(\frac{C}{m} \right)$$

Surface charge density:



conducting sphere.

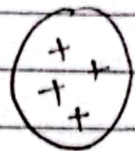


⇒ Also called surface charge density:

⇒ Charge per unit area is called surface charge density

$$\sigma = \frac{q}{A} \quad \left(\frac{C}{m^2} \right)$$

3) Volume Charge Distribution:



non-conducting sphere.

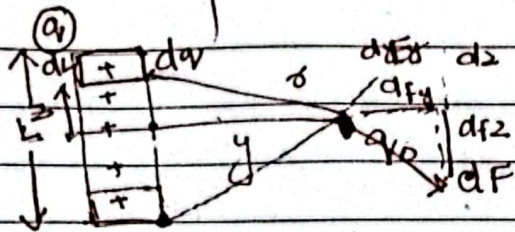
→ Volume charge density

→ Charge per unit volume

$$\rho = \frac{q}{V} \left(\frac{C}{m^3} \right)$$

Linear Charge distribution:

→ A uniform line of Charge (Linear charge distribution)



$$dz = 0$$

$$dx = 0$$

→ Consider a thin rod length $-\frac{L}{2}$ to $+\frac{L}{2}$, that

carries a uniformly distributed +ve charge (q)

So the linear charge density will be

$$\lambda = \frac{q}{L}$$

We want to calculate force exerted by rod

on a +ve point charge (q_0) placed at perpend

Bisector of rod, at distance of 'y' from its

Centre:

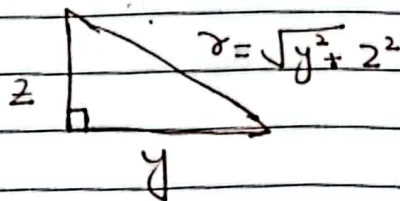
We divide the rod into small elements of

length dz .

An arbitrary charge which is at distance z from q_0 will exert a force on q_0 which can be written as:

$$dF = \frac{K q_0 dq}{r^2}$$

$$dF_y = dF \cos \theta \rightarrow (B)$$



$$dF_y = \frac{K q_0 dq}{r^2} \left(\frac{y}{r} \right) \left(\frac{8\pi \times 10^9 \text{ N m}^2/\text{C}^2}{9} \right) \left(\lambda = \frac{dq}{dz} \right)$$

($dq = \lambda dz$)

$$dF_y = \frac{K q_0 (\lambda dz)}{r^2} \left(\frac{y}{r} \right)$$

$$dF_y = \frac{K q_0 (\lambda dz) y}{r^3} \rightarrow (C)$$

$$\int dF_y = \int K q_0 \lambda y \left(\frac{dz}{r^3} \right)$$

$$F_y = K q_0 \lambda y \int_{-L/2}^{+L/2} \frac{dz}{(y^2 + z^2)^{3/2}} \rightarrow (D)$$

$$\therefore \int_{-\frac{L}{2}}^{+\frac{L}{2}} \frac{dz}{(y^2+z^2)^{3/2}}$$

$$z = y \tan \theta$$

$$\boxed{dz = y \sec^2 \theta d\theta}$$

$$-\frac{L}{2} = y \tan \theta$$

$$-\frac{L}{2y} = \tan \theta$$

$$\theta = \tan^{-1} \left(-\frac{L}{2y} \right)$$

$$\theta = -\tan^{-1} \left(\frac{L}{2y} \right)$$

$$\theta = +\tan^{-1} \left(\frac{L}{2y} \right)$$

$$\tan^{-1} \left(\frac{L}{2y} \right) = +n$$

$$y \sec^2 \theta$$

$$(y^2 + \underbrace{y^2 \tan^2 \theta}_z)^{3/2}$$

$$-\tan^{-1} \left(\frac{L}{2y} \right) = -n$$

$$+n$$

$$y \sec^2 \theta$$

$$y^{3/2} (1 + \tan^2 \theta)^{3/2}$$

$$\sec^2 \theta$$

$$(1 + \tan^2 \theta)^{3/2}$$

$$= \int_{-n}^{+n} \frac{\sec^2 \theta}{y^2 \sec^3 \theta}$$

$$\frac{1}{y^2} \int \frac{1}{\sec \theta} d\theta$$

$$\frac{1}{y^2} \int \cos \theta d\theta$$

$$\frac{1}{y^2} \sin \theta \Big|_{-n}^{+n}$$

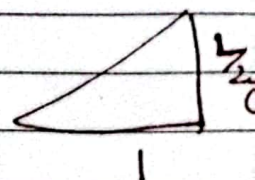
sin

$$\frac{1}{y^2} \sin(n) - \sin(-n)$$

$$\frac{1}{y^2} 2(\sin n)$$

$$\frac{2}{y^2} \sin n$$

$$\frac{2}{y^2} \sin \left(\tan^{-1} \frac{L}{2y} \right)$$



$$\tan = \frac{L}{2y}$$

$$H = \sqrt{B^2 + p^2}$$

$$\sqrt{1 + \frac{L^2}{4y^2}}$$

(integration)

$$\sin \alpha = \frac{p}{H}$$

$$\sin \alpha = \frac{L/2y}{\sqrt{1 + L^2/4y^2}}$$

$$\int \frac{2}{y^2} \sin(\alpha)$$

$$\int \frac{2}{y^2} \left(\frac{p}{H} \right)$$

$$\left[\frac{2}{y^2} = \frac{L/2y}{\sqrt{1 + L^2/4y^2}} \right]$$

$$\frac{L}{y^3} \times \frac{1}{\sqrt{1 + \frac{L^2}{4y^2}}}$$

$$\frac{L}{y^3} \times \frac{1}{\sqrt{\frac{1}{y^2} (y^2 + \frac{L^2}{4})}}$$

$$\frac{1}{y^3} \times \frac{1}{\sqrt{\frac{1}{y} (y^2 + \frac{L^2}{4})}} \quad \text{--- (E)}$$

$$F_y = (Kq_0)(\lambda y) \times \frac{1}{y^2} \times \frac{1}{\sqrt{y^2 + \frac{L^2}{4}}}$$

$$F_y = \underbrace{(K q_0)}_4 \times \underbrace{(a y)}_{y^2} \times \underbrace{1}_{\sqrt{y^2 + \frac{L^2}{4}}}$$

$$F_y = \frac{K q_0 a}{y \sqrt{y^2 + \frac{L^2}{4}}}$$

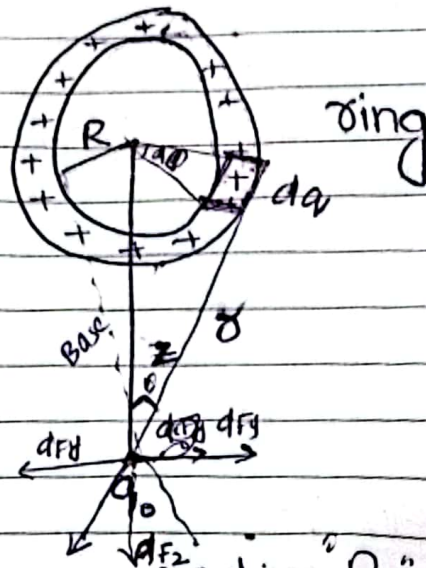
$$F_y = \frac{1}{4\pi\epsilon_0} \times \frac{q_0 a}{y \sqrt{y^2 + \frac{L^2}{4}}}$$

Electric field

$$F_y = \frac{F}{q_0} = \frac{K a}{y \sqrt{y^2 + \frac{L^2}{4}}} \left[\frac{1}{4\pi\epsilon_0} = K \right]$$

$$E_y = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{a}{y \sqrt{y^2 + \frac{L^2}{4}}}$$

Electric Force due to ring of charge



length Formula

$$S = R\theta$$

$$S = R d\phi$$

Consider a thin ring of radius R carries a uniformly distributed charge q so that linear charge density will be given as

$$\lambda = \frac{q}{\text{length}}$$

$$\lambda = \frac{q}{2\pi R}$$

$$\lambda = \frac{q}{\text{Circumf. of circle}}$$

⇒ We want to calculate force exerted by Ring on a point Charge (+ve) q_0 along z -axis at a distance z from the centre of the ring.

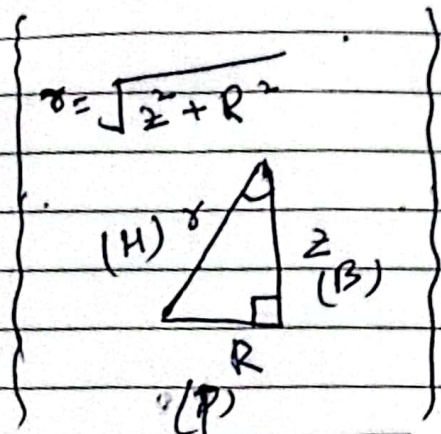
⇒ Consider a small element of length $(R d\phi)$ having charge q dq

$$\lambda = \frac{dq}{R d\phi}$$

$$\left\{ \begin{array}{l} S = R \theta \rightarrow \text{Formula} \\ S = R d\phi \end{array} \right\}$$

$$dq = (\lambda)(R d\phi)$$

$$\star \rightarrow dF = \frac{kq_0 da}{r^2}$$



$$dF_2 = dF \cdot \cos \theta$$

Put value of dF

$$dF_2 = \frac{kq_0 da}{r^2} \left(\frac{B}{H} \right)$$

$$\left\{ \cos \theta = \frac{B}{H} \right\}$$

$$dF_2 = \frac{kq_0 da}{r^2} \left(\frac{z}{r} \right)$$

$$dF_2 = \frac{kq_0 da z}{r^3}$$

Put the value of da

$$dF_2 = \frac{kq_0 (\lambda R d\phi) z}{r^3}$$

$$dF_2 = \frac{kq_0 (R \lambda d\phi) z}{(R^2 + z^2)^{3/2}}$$

$$dF_2 = \frac{kq_0 \lambda R z}{(z^2 + R^2)^{3/2}} d\phi$$

integrate it.

$$\int dF_2 = \frac{kq_0 \lambda R z}{(z^2 + R^2)^{3/2}} \int d\phi$$

$$F_2 = \frac{kq_0 \lambda R z}{(z^2 + R^2)^{3/2}} \int_0^{2\pi} d\phi$$

$$F_2 = \frac{kq_0 \lambda R z}{(z^2 + R^2)^{3/2}} \left\{ \phi \Big|_0^{2\pi} \right\}$$

$(2\pi) - (0) = 2\pi$

$$F_2 = \frac{kq_0 \lambda R z}{(z^2 + R^2)^{3/2}} (2\pi)$$

$$F_2 = \frac{kq_0 \left(\frac{Q}{2\pi R} \right) \times (2\pi R) z}{(z^2 + R^2)^{3/2}}$$

$$F_2 = \frac{kq_0 Q z}{(z^2 + R^2)^{3/2}}$$

The final force exerted by circle on charge q_0

Special cases:

$z \gg R$

(distance more greater than size and point charge)

$$F_z = \frac{K q_0 q z}{(z^2 + R^2)^{3/2}}$$

We can neglect R : $R \approx 0$

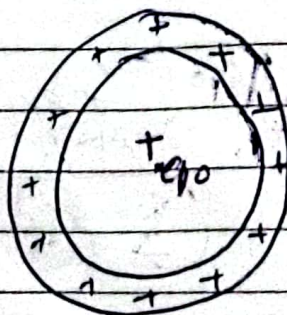
$$F_z = \frac{K q_0 q z}{(z^2 + 0)^{3/2}}$$

$$F_z = \frac{K q_0 q z}{z^3}$$

$$F_z = \frac{K q_0 q}{z^2}$$

↳ This is Coulombic force means if distance is more then it will behave as single Coulombic force.

Case 2:



$$F_z = 0$$

$$F_z = 0$$

Because all the centre of ring, charge will be equally distributed in all directions

مطلب net force 0

کیونکہ ہر چیز اساتر ہے

سب گیلی

but the charge is

Electric Field:-

A region or space around a charge in which it can exert electrostatic force of attraction or repulsion on another charge.

Electric field intensity:



(divide the force on point charge)

If a small charge q_0 will experience force ' F ' in electric field ' E ', then electric field intensity is given as

$$\vec{E} = \frac{\vec{F}}{q_0}$$

→ It is a vector quantity

→ its unit is N/C

→ It is always in the direction of force

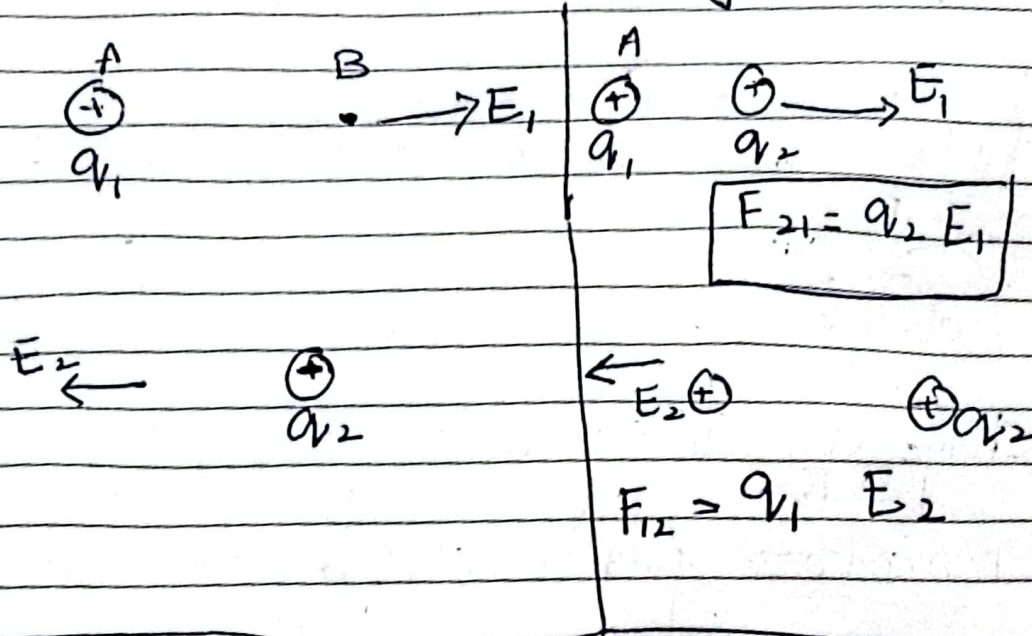
→ q_0 must be small enough so that, it will not disturb the field produced by source of charge

So we can write E as

$$E = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$$

Charge basically doesn't interact directly
it is as:

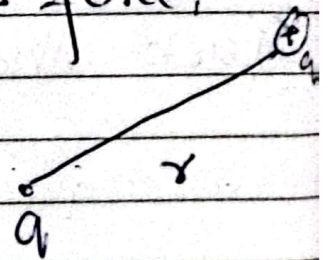
Charge $\rightleftharpoons$ Field $\rightleftharpoons$ Charge



Electric field due to point Charge:

Let us consider a (+ve) test charge at a distance ' r ' from a point charge ' q ', we can calculate the magnitude of Coulombs force,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_0 [q]}{r^2}$$



Similarly the magnitude of electric field at site of test charge is given as

$$E = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} \quad \text{--- (1)}$$

→ $\vec{F}$ and $\vec{E}$ will point out if q is +ve.

→ $\vec{F}$ and $\vec{E}$ will point inward if q is negative

→ Electric field vector form is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Electric field due to N-charges

$N = \text{many}$

Let us consider we have $q_1, q_2, q_3, \dots, q_n$ Charges

and related electric fields $E_1, E_2, E_3, \dots, E_n$

We want to Calculate Electric field due to

Group of N-charges, we will take the following procedures:

① Calculate E_n due to each n charge assuming absence of other Charges

② Calculate total ' E ' by taking vector sum of intensities of individual charges.

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots + \vec{E}_n$$

$$\sum \vec{E} = \sum \vec{E}_n$$

This is an application of principle of superposition.

Electric field due to Dipole

a pair of dipole of $\pm$ magnitudes & distance b/w $\pm$ -ve & +ve

Dipole:

① A pair of (+ve) and (-ve) charge having same magnitude and constant distance b/w them is called dipole.

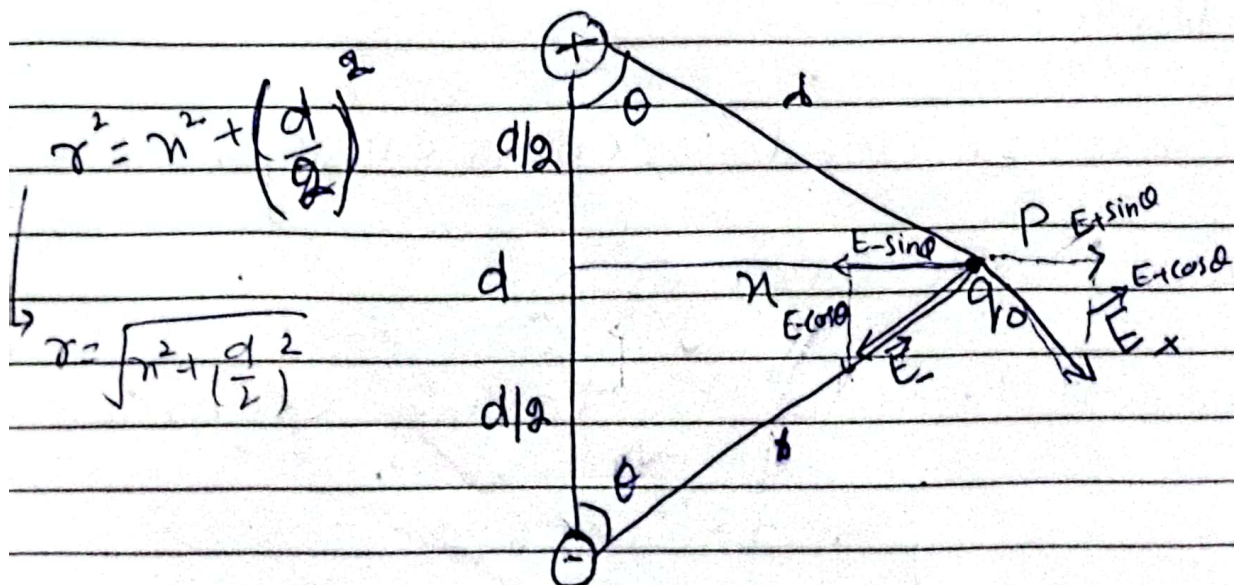
Dipole moment:

Product of charge and separation b/w them is called dipole moment

$$P = qd$$

→ its a vector quantity.

its direction always from -ve to +ve charge.



→ Consider a positive and negative charge have same magnitude and separation b/w them is 'd', they constitute a dipole & we want to calculate the electric field at point P due to dipole.

→ Draw a perpendicular Bisector of line joining charges and consider point 'P' is at distance 'r' from line. Since point 'P' is at equal distance from positive and negative charges, so magnitude of E_+ and E_- will be same;

$$|E_+| = |E_-|$$

$$E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q_0}{r^2}$$

→ total E

As electric field follows principle of superposition so

$$\boxed{E = E_+ + E_-}$$

Now we want value of E_+ & E_-

ix Component of E:

As $E_+ \sin\theta$ and $E_- \sin\theta$ are equal in magnitude but opposite in direction, so they will cancel each other

$$E = E_+ \sin \theta - E_- \sin \theta$$

$$= E_+ \sin \theta - E_- \sin \theta$$

$$E_+ = E_-$$

$$E = E_+ \sin \theta - E_+ \sin \theta = 0$$

Z. components of E :

$$E = E_+ \cos \theta + E_- \cos \theta$$

$$E_+ = E_- \text{ so}$$

$$E = E_+ \cos \theta + E_+ \cos \theta$$

$$E = 2 E_+ \cos \theta$$

$$E_+ = E_- = E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{d}{2r} \right)$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^3}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{dq}{\left(r^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{\left(r^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}} \quad \text{Electric field due to dipole:}$$

Case:

if we take r^2 as common;

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3 \left[1 + \left(\frac{d}{2r} \right)^2 \right]^{3/2}}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3} \left[1 + \left(\frac{d}{2r} \right)^2 \right]^{-3/2}$$

Binomial Expansion:

$$(1+x)^n = (1+nx + \dots)$$

Applying binomial Expansion:

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3} \left[1 + \left(\frac{-3}{2} \right) \left(\frac{d}{2r} \right)^2 + \dots \right]$$

if $r \gg d \Rightarrow d=0$

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3}$$

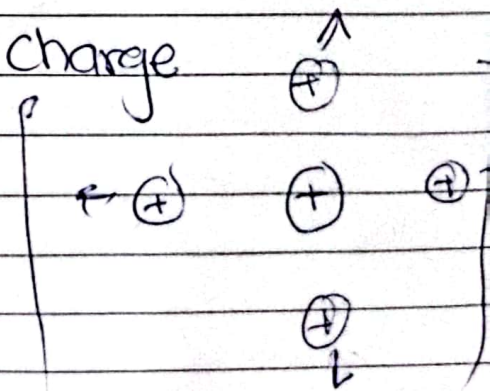
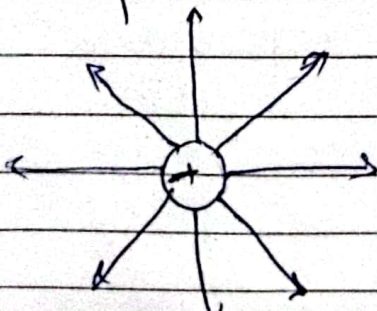
Lecture: 15-October:

Electric field intensity: Lines:

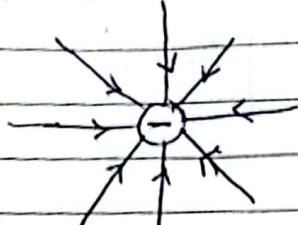
$$E = \frac{F}{q} \quad \text{OR} \quad E = k \frac{q}{r^2}$$

Electric field lines are drawn to specify the direction of electric field:

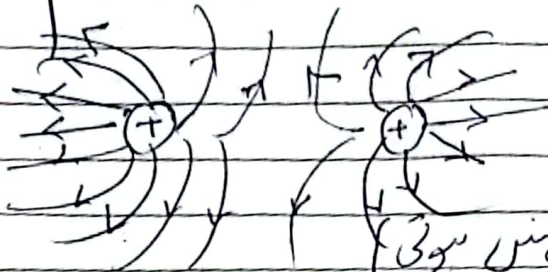
→ electric field lines due to +ve charge



→ Electric field lines due to -ve charge:

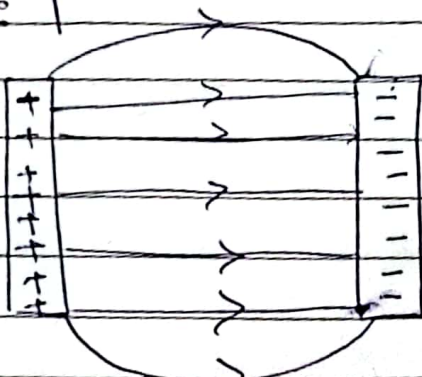


→ Electric field lines due to same +ve charges:



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→ Electric field lines b/w two oppositely charged plates:



→ Properties of electric field lines.

① → electric field lines originate from the charge and terminate into -ve charge:

② → Electric field lines never cross each other:

③ → which means they have only one direction at given

③ point

→ Electric field lines are close to each other where electric field is strong and are further apart where the field is weak:

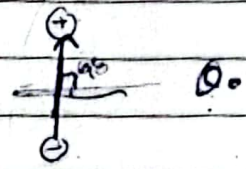
3) Tangent to field lines gives +ve direction to electric force

3) Electric field lines are imaginary but electric field is real.

Dipole in an electric field:

A pair of +ve and negative charges same

in magnitude and constant distance is called

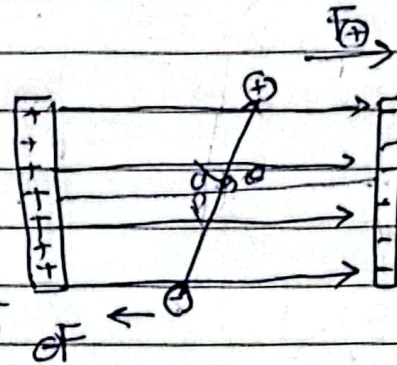


dipole:

Let's consider this dipole

is placed in an external

electric field (E) such that



dipole moment vector (P) makes an angle (θ) with (E)

$$|H| = |F| = F$$

A force positive ($+F$) will be exerted on (+ve) and

$-F$ on $(-ve)$. As the force equal in magnitude but opposite

direction so they will produce a couple

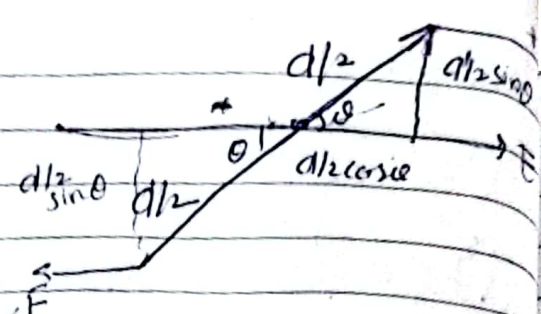
(because they are non-co-linear means equal in magnitude

opposite direction but not on same line).

And this couple will tend to rotate electric dipole vector

($\vec{P}$) around its center of mass.

Torque:



$$\tau = \tau_1 + \tau_2$$

$$\tau = \left(\frac{d}{2} \sin \theta\right) + \left(\frac{d}{2} \sin \theta\right) F$$

$$= F \sin \theta \left(\frac{d}{2} + \frac{d}{2}\right)$$

$$= F \sin \theta \left(\frac{2d}{2}\right)$$

$$= F \sin \theta d = \boxed{\tau = F d \sin \theta} \checkmark$$

Force = qE So

$$\tau = (qE) d \sin \theta$$

$$\tau = (qd) E \sin \theta$$

$$\tau = p E \sin \theta$$

$$\boxed{\vec{\tau} = \vec{p} \times \vec{E}}$$

Direction of torque will be
the into plane of board

WORK-DONE

$$W = Fd$$

$$dw = F \cdot ds \text{ (small work)}$$

$$dw = Fd (\cancel{r} d\theta)$$

$$dw = r F \cdot d\theta$$

$$dw = \tau \cdot d\theta$$

By integrating:

$$\int_{\theta_1}^{\theta_2} dw = \int_{\theta_1}^{\theta_2} \tau d\theta$$

$$\cancel{s = r\theta}$$

distance = linear displacement

Linear work
 $W = F \cdot d$
angular work

$$dw = \tau \cdot d\theta$$

→ because angle reduced from this

$$W = - \int_{\theta_0}^{\theta} \tau d\theta$$

$$W = - \int_{\theta_0}^{\theta} PE \sin \theta d\theta$$

$$W = - PE \int_{\theta_0}^{\theta} \sin \theta d\theta$$

$$W = - PE (-\cos \theta) \Big|_{\theta_0}^{\theta}$$

Integration of $\sin \theta = -\cos \theta$

Put limits

$$W = PE (\cos \theta - \cos \theta_0)$$

Potential Energy:

$$W = -\Delta U$$

$$\Delta U = -W$$

$$\Delta U = -PE (\cos \theta - \cos \theta_0)$$

$\theta_0 = \text{initial angle} = 90^\circ$

$$U(\theta) - U(\theta_0) = -PE (\cos \theta - \cos 90^\circ)$$

$$U(\theta) = -PE \cos \theta$$

Special cases

1) Maximum energy: (PE)

$$U = -PE \cos \theta$$

$$\theta = 180^\circ$$

$$\cos 180 = -1$$

$$U = -PE (-1)$$

$$U_{\max} = PE$$

2) Minimum energy (P.E) when P and E are parallel

$$\theta = 0^\circ$$

$$\cos 0^\circ = 1$$

$$U = -PE (1)$$

$$U_{\min} = -PE$$

Electric Flux:

Electric field lines passing through an area held perpendicular to the direction of electric field

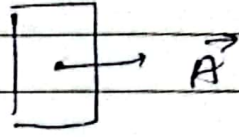
→ It is dot product of electric field $\vec{E}$ and vector area $\vec{A}$

$$\phi = \vec{E} \cdot \vec{A}$$

(vector area)

(magnitude = surface area)

(direction → perp to surface area)



$$\phi = EA \cos \theta$$

if perpendicular object (مستقيم)
vector area (المنطقة المتجهة)

→ It is a scalar quantity

$$\text{Unit} = \text{NC}^{-1} \text{m}^2 \Rightarrow \frac{\text{N}}{\text{C}} \text{m}^2$$

vector (متجه)
scalar (سكالر)

Case 1:

When surface area is held perpendicular to E.F line

$$\phi = \vec{E} \cdot \vec{A}$$

$$\phi = EA \cos 0$$

$$\phi = EA \cos 0^\circ$$

$$\phi = EA (1)$$

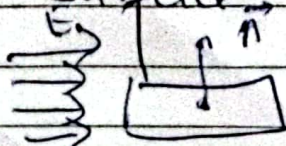
$$\phi = EA$$

$$\phi_{\text{max}} = EA \text{ (maximum)}$$

In this case ϕ is maximum

Case 2:

When surface area is parallel to E.F line



$$\phi = \vec{E} \cdot \vec{A}$$

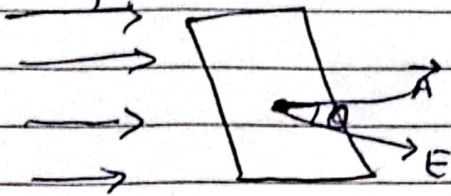
$$\phi = EA \cos 90^\circ$$

$$\phi = 0$$

In this case, flux will be minimum:

Case 3:

When surface area is neither parallel nor perpendicular



$$\phi = \vec{E} \cdot \vec{A}$$

$$\phi = EA \cos \theta$$

In this case the flux will neither be minimum nor maximum.

Gauss Law

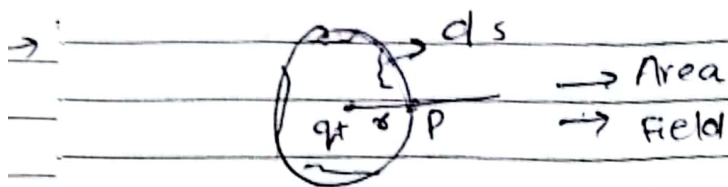
Electric flux through a close surface is equal to $\frac{1}{\epsilon_0}$ times to the total charge enclosed by the surface.

$$\phi = \frac{1}{\epsilon_0} \times q_{\text{enclosed}}$$

$$\phi = \frac{q_{\text{enclosed}}}{\epsilon_0} \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ C/Vm}^2$$

Explanation:

Let us consider a sphere of radius r having a charge $(+q)$ at its centre.



→ Electric field at a point 'P' be written as,

$$E = \frac{kq}{r^2} \quad E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

→ Total flux through sphere

$$\phi = \vec{E} \cdot \vec{ds}$$

$$\phi = \oint \vec{E} \cdot \vec{ds} \quad \left(\oint \rightarrow \text{close surface} \right)$$

$$\phi = \oint E ds \cos \theta$$

$$\phi = \oint E ds \cos 0^\circ$$

$$\phi = \oint E ds$$

$$\phi = E \oint ds$$

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \oint ds$$

$\left[\oint ds \rightarrow \text{total surface area of sphere} \right]$

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \times 4\pi r^2$$

$$\boxed{\phi = \frac{q}{\epsilon_0}} \quad \text{hence proved}$$

$$\phi = \frac{1}{\epsilon_0} \times q_{\text{enclosed}}$$

Application of Gauss's Law:

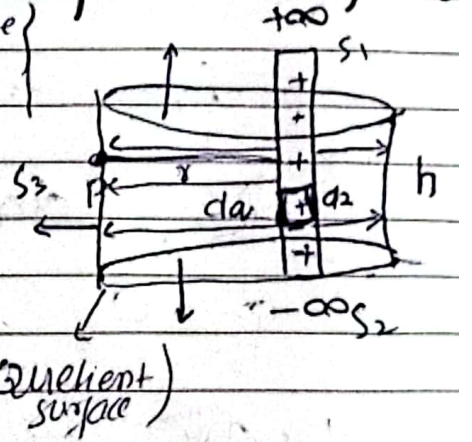
1) Electric field intensity due to infinite line of charge

→ Let us consider a rod of $\begin{cases} dq = \text{charge} \\ dz = \text{length} \end{cases}$

infinite length, having a

uniformly distributed (+ve q)

charge over it.



→ We want to calculate E.F.I at a point 'P' which is at distance ' r ' from the rod.

→ Let us draw an imaginary Gaussian surface of in the form of cylinder of length ' h '

→ Consider a small charge dq such that linear charge density

$$\lambda = \frac{dq}{dz}$$

$dq = \lambda dz$ → linear charge density for small charge.

integrate for total length charge

$$dq = \lambda dz$$

$$q = \lambda h$$

$$\left\{ \text{where } dz = h \right\}$$

→ Total flux through Gaussian surface

$$\phi = \int \vec{E} \cdot d\vec{a}$$

$$\phi = \int E da \cos \theta$$

→ Flux from S_1 (Top)

$$\phi = \int E da \cos 90^\circ$$

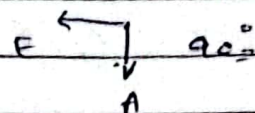
$$\phi_1 = 0$$

→ Flux from S_2 (Bottom)

$$\phi = \int E da \cos \theta$$

$$\phi_2 = \int E da \cos 90^\circ$$

$$\phi_2 = 0$$

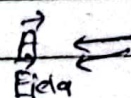


→ Flux from S_3 (middle)

$$\phi = \int E da \cos \theta$$

$$\phi_3 = \int E da \cos 0^\circ$$

$$\phi_3 = \int E da$$



$$\phi_3 = E \int da$$

$$\phi = E(2\pi r)h$$

Total "

$$\phi = \phi_1 + \phi_2 + \phi_3 = 0 + 0 + E(2\pi r)h$$

$$\phi_{\text{total}} = E(2\pi r)h \quad \text{--- (A)}$$

→ Calculate flux by Gauss's law.

$$\phi = \frac{q}{\epsilon_0} \quad \text{--- (B)}$$

Comparing both eq (A) (B)

$$E(2\pi r)h = \frac{q}{\epsilon_0}$$

$$E(2\pi r)h = \frac{\lambda h}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi r \epsilon_0}$$

So E depends on $\frac{\lambda}{r}$

ii) **Electric field intensity due to infinite Sheet of charge:**

→ Consider a positive charge

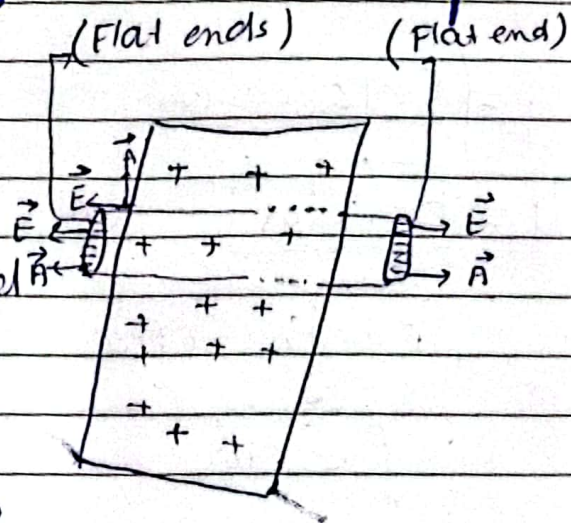
(+ve) is uniformly distributed

over a sheet of infinite

extent such that surface

charge density

$$\sigma = \frac{q}{A} \Rightarrow q = \sigma \cdot A$$



In order to calculate electric field intensity at

Point 'P', imagine a gaussian surface in form of cylinder such that its one end contains

→ Point at which $E \cdot F$ is to be calculated.

→ Total flux is

→ Flux to flat surface (left)

$$\phi = \int \vec{E} \cdot d\vec{a}$$

$$\phi = \int E da \cos \theta$$

$$\phi = \int E da$$

$$\boxed{\phi_1 = E \int da}$$

Flux through flat surface (Right)

$$\phi = \int \vec{E} \cdot d\vec{a}$$

$$\phi = \int E da \cos \theta$$

$$\boxed{\phi_2 = E \int da}$$

Flux through cube surface

$$\phi = \int \vec{E} \cdot d\vec{a}$$

$$\phi = \int E da \cos \theta$$

$$\phi = \int E da \cos 90^\circ$$

$$\phi = E \int da \cos 90^\circ$$

$$\boxed{\phi_3 = 0}$$

$$\phi = \phi_1 + \phi_2 + \phi_3 \Rightarrow E \int da + E \int da + 0 = 2E \int da$$

$$\phi = 2EA \text{ — (A)}$$

$$\left[\int da = A \right]$$

→ Flux by using Gauss's law:

$$\phi = \frac{q}{\epsilon_0}$$

$$\phi = \frac{dA}{\epsilon_0} \text{ — (B)}$$

$$2EA = \frac{dA}{\epsilon_0}$$

$$E = \frac{d}{2\epsilon_0}$$

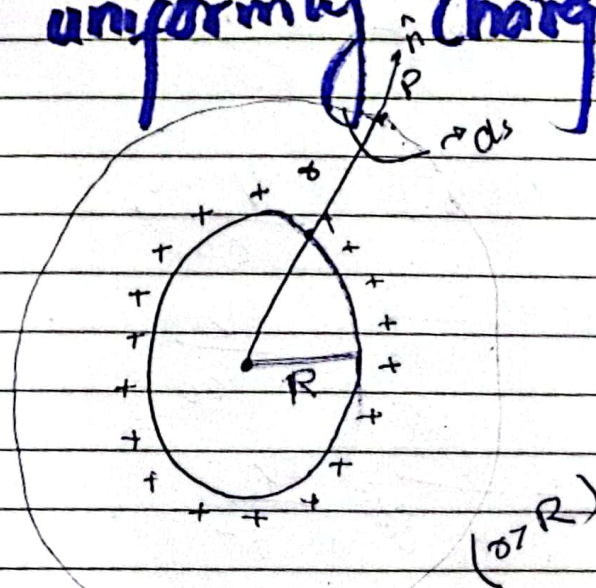
Electric field due to infinite sheet of charge

$$\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$$

→ Vector form:

3) Electric field due to uniformly charged Shell:

Let us consider a +ve charge (+q) is uniformly distributed over a shell of radius 'R'



→ Let's calculate electric field at different points.

1) Outside the shell:

→ Outside the shell:

→ In order to calculate the E.F at a point 'P' imagine

a surface of radius 'r' such

$$\left[\sigma = \frac{q}{A} \right]$$

charge density

→ Indt ($r > R$)

→ Flux is

$$\phi = \oint \vec{E} \cdot d\vec{s}$$

$$\phi = E \int ds \cos \theta$$

$$\phi = E \int ds \cos 0^\circ$$

$$\boxed{\phi = E \int ds}$$

Electric field will be directed radially outward and will be parallel to unit vector $\hat{r}$
So $\theta = 0^\circ$

→ Flux by Gauss law

$$\phi = \frac{q_{in}}{\epsilon_0}$$

$$E \int ds = \frac{q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{q_{in}}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_{in}}{r^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

→ A uniform spherical shell of charge behaves as all charge is concentrated at its centre

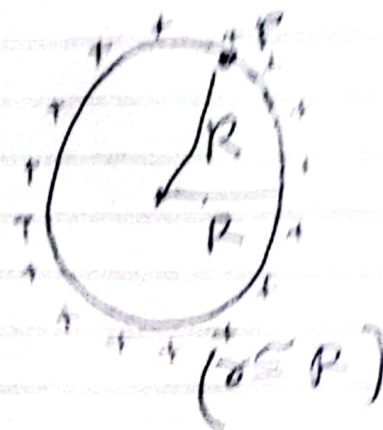
(It is called First Shell Theorem)

($E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \Rightarrow$ Electric field due to a point charge)

2) at surface of shell:

$$(r = R)$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$



3) inside the shell:

$$(r < R)$$

As the charge is uniformly distributed over the surface

of shell, so there is no charge inside

the shell so $q_{in} = 0$



$$\int \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$$

$$\int E ds \cos \theta = \frac{0}{\epsilon_0}$$

$$\int E ds \cos \theta = 0$$

$$E \int ds \cos \theta = 0$$

$$E = 0$$

[Force will also be zero $\Rightarrow F = qE$]

A uniform spherical shell of charge will exert no any force on a charge particle placed inside the shell $\rightarrow$ (Also called 2nd shell theorem)

Assignment 1

25.2	26.1
25.3	26.2
25.4	26.3
	26.4

problem 1

Quiz

From start to guass law